BIOCHEMISTRY CUMULATIVE EXAM

December 2, 1995

Cytochrome oxidase is an integral mitochondrial membrane protein which transfers four electrons from four molecules of cytochrome c to O_2 to form two molecules of water. Cytochrome c has a redox potential of $E_o' = +0.26$ V, and O_2 has a redox potenital of $E_o' = +0.82$ V. Under active electron transfer conditions, the electrical potential across the mitochondrial membrane was measured to be $V_{out} - V_{in} = +0.1$ V, and the pH inside and outside the mitochondrial membrane was $pH_{in} = 7$, and $pH_{out} = 6$.

1. (25) Give a simple diagram showing how cytochrome oxidase couples the energy of electron transfer to pumping protons across the mitochondrial membrane, giving the total number of protons pumped per O_2 consumed.

2. (25) Calculate the free energy change ΔG^o for electron transfer from 4 molecules of cytochrome c to one molecule of O_2 to form two H_2O.

3. (25) Give a formula for the free energy change ΔG for pumping protons from the inside to the outside of the mitochondrial membrane, and calculate ΔG for the proton pumping component of the cytochrome oxidase reaction under the above conditions.

4. (25) Discuss how efficiently cytochrome oxidase couples the exergonic free energy of electron transfer to the endergonic free energy of proton pumping under the above conditions.

Possibly useful formulas:

$\Delta G^o = - RT\ln K_{eq}$; $\Delta G^o = -nF\Delta E_o$;
Membrane Transport: $\Delta G = RT\ln\{c_2/c_1\} + ZF\Delta V$
F = Faraday's constant = 23 kcal $volt^{-1}$ mol^{-1} ; Z = charge
$R = 1.987$ cal mol^{-1} deg^{-1}